Space, Matter and the Clock Do Not Share a Common Birth Condition — What a Seed Must Be to Make a Particle, and the Lower Bound of Resolution

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What kind of paper this is

This paper does not propose a new dynamics. It imports, read-only and without any modification, the dynamics established in the previous paper, and changes only **where the seed is placed, how strong it is, its phase, the resolution, and the number of divisions of the period**, then measures what happens.

Every number quoted here was recomputed independently from the stored run data and checked mechanically against the existing records (Appendix E). Where an experiment was run after its prediction had been fixed in a document, this is stated in the text.

Premise

In the previous paper, 62 wave shapes were identified as candidates corresponding to the particles of the Standard Model. In the course of that work it was found that, in the inflation experiment, waves appearing at odd harmonics behave in a fermionic way and waves at even harmonics behave in a bosonic way; and that when the reflection (mixing) ratio of the interaction is around 0.7, a state mixing the fermionic and the bosonic appears. The mechanism behind that mixing was left open.

In particular, geometric-series expansion of the inflationary kind occurred **without** any seed, whereas a seed was **indispensable** for particle-like waves to appear. Only one kind of seed, of comparatively large amplitude, had been tried, so which seed produces which particle-like wave was also left open.

The influence of the resolution N was likewise left unanalysed.

This paper addresses those questions and obtains definite results.

The results, sorted into five levels of confidence

So that the reader does not confuse them, the findings are classified up front.

(1) Strong measured regularities

- The onset of expansion is governed by the logarithm of the seed's **complex sum** ($R^2 = 0.99987$)

- The time at which particle-like waves finish appearing is governed by a power of the **power placed on odd bands** ($R^2 = 0.996$, 14 points)
- With a seed on even bands only, the odd bands stay **exactly zero** (42000 updates × 8 strengths)
- The classification by greatest common divisor **follows the number of divisions of the period** (278 pairs, zero failures)

(2) Understood down to the mechanism

- The onset depends on the complex sum because the first half of one update looks only at that sum and applies the same rotation to every location
- Nothing happens with an even-band-only seed because the coefficient of the interaction is built from the ratio of odd-band to even-band power; if the odd bands are zero, the coefficient is zero. That the odd bands never appear also follows, as parity conservation, from the three-wave form of the interaction

(3) Empirical only

- The exponent 1.073 itself (why it is close to 1 is not explained)
- That selectivity toward the targeted location peaks at intermediate strength
- The differences of regime with resolution
- Whether the first half of an update reads the **sum** or the **sum of squares** has not been separated experimentally (the update procedure says it is the sum)

(4) Recorded as anomalies, interpretation withheld

- At resolution 4 the clock runs 9.9 times faster
- At resolution 4 only, the readout crosses the value 0.6972
- At resolutions 8 and 10 the initial state could not be built
- At resolutions 13 and 20 the competition between planes reappears

(5) Correspondence hypotheses to real physics (not tested here)

- That odd and even bands correspond to fermions and bosons
- That the birth of a third direction corresponds to the birth of real space
- That the clock the system carries corresponds to physical time
- That a seed corresponds to a source of particle production

None of (5) is tested in this paper; all of it is inherited from the previous paper. What is measured here is only which quantities become readable, and under which conditions, inside the model.

How the measurements were conducted

What is most often doubted in independent research is whether the conditions were chosen after seeing the results. The following procedure was used.

- **Some experiments had their predictions fixed in a document before being run.** The four conditions of the phase-cancelled seed (section 3) and the test of changing the number of divisions of the period (section 8) were run after the predicted values had been written down, and hit as predicted
- **The dynamics is imported read-only, and checks its own digest before and after each run.** A mismatch stops the run on the spot
- **Every number in the claims was recomputed from the stored data and checked against the existing records** (Appendix E)
- **All 462 per-run record figures were inspected.** As a result, six of the claims in the text were found to need correction, and were corrected. In particular, the “window of strength” in section 2 was found by this inspection to be an artefact of the observation time
- Digests of 126 runs, 1174 MB of stored data and 13 programs are listed in the appendix

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The experimental system

Only what is needed to read the claims is given here.

Points, relations, resolution

The system consists of **N points**. Every pair of points carries a **relation**, so there are N(N1)/2 relations. This N is called the **resolution**. The main experiment uses N = 12, hence 66 relations.

Waves and "locations"

Each relation carries one wave, but that wave is not a single complex number. It is a set of values arranged along two periodic directions. The first direction is divided into 16 parts of one turn, the second into 8, so there are 128 values in all.

A wave along a periodic direction can be classified by how many times its phase turns in one lap. That count is called the **winding number**. Write k for the winding number of the first direction and m for that of the second. The 128 values can be separated into components indexed by the pair (k, m). In this paper a **location** means such a pair. There are 128 locations.

A location is not an address that the wave remembers. Because the wave is a distribution rather than a single complex number, its components simply carry indices.

Locations sharing the same k are grouped into a **band**; there are 16 bands. The previous paper found that **waves on odd bands behave in a fermionic way and waves on even bands in a bosonic way**, and this paper inherits that correspondence unchanged.

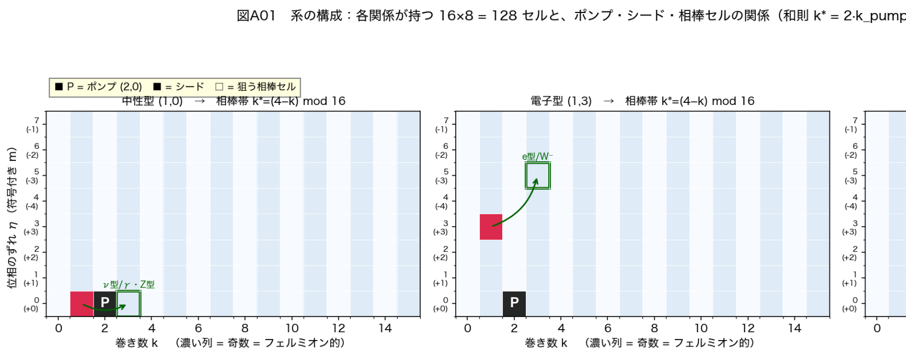

Figure 1The system. The map of 128 locations carried by each relation, and the relation between the background oscillation (black), the seed (red) and the location it targets (green frame). Dark columns are odd bands.

The background oscillation and the seed

Before a run starts, a wave of large amplitude is placed at location (k = 2, m = 0). This is the **background oscillation**. It is common to every condition, including the "no seed" case below. **It is because of this background oscillation that inflation-like expansion occurs even without a seed.**

On top of that, adding a small amplitude at specified locations is called **seeding**. The magnitude added is the **seed strength**, and the set of locations is the **seed placement**.

Updates, and the count kept from outside

The system repeats discrete updates. **There are no seconds in this paper; all time is measured in numbers of updates.** That count is a scale we keep from outside; it is not inside the system.

One update has two halves. The first forms, for each relation, the **sum of all 128 values taken as complex numbers**, decides a rotation from the argument of that sum, and **applies the same rotation**

to all 128 locations. The second half is the interaction in which three waves meet and make another.

図A03　代表的な通し走行（混合シード・δ=0.01・N=12・42000 回）——三つの誕生が一本の走行の中で順に起こる

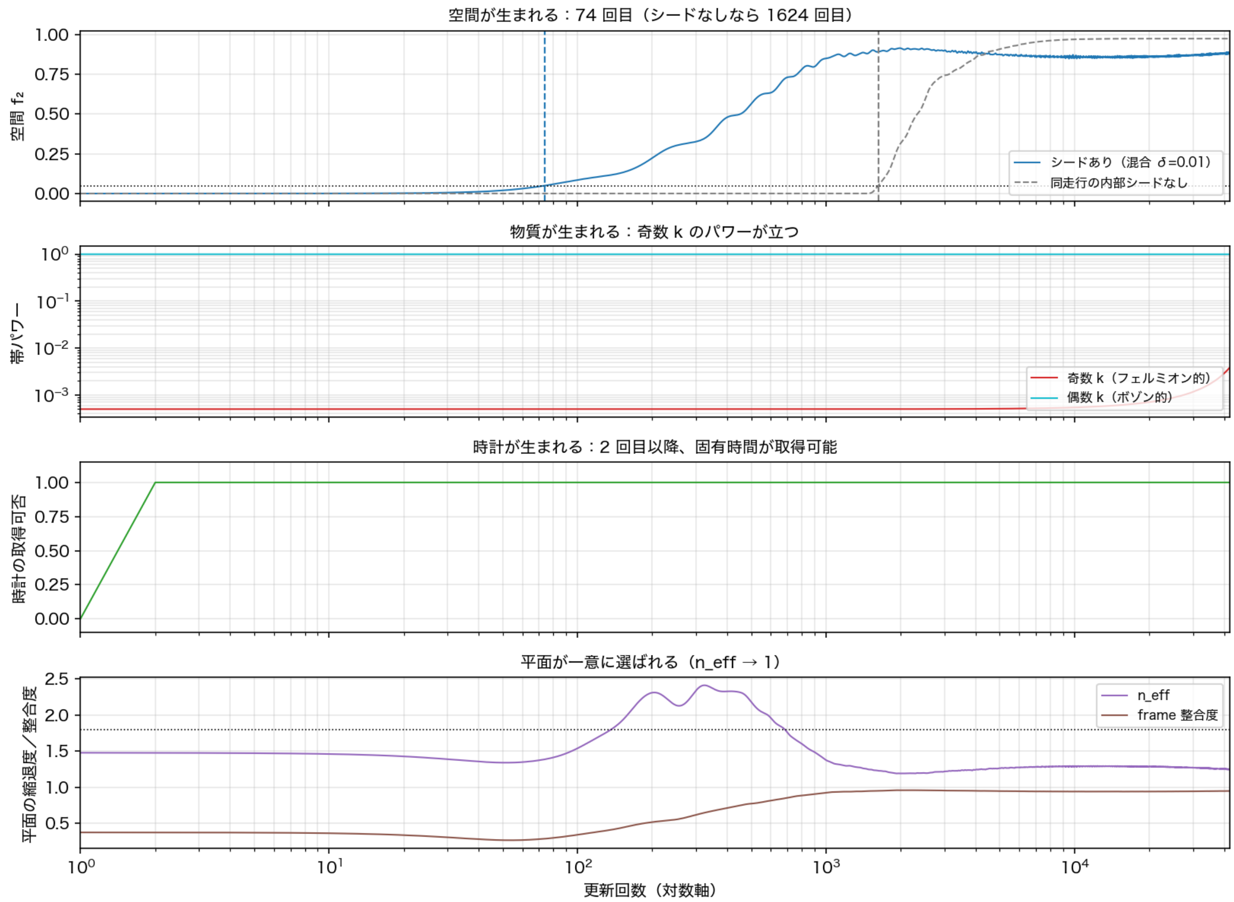

Figure 2One representative run (mixed 8 locations, strength 0.01, resolution 12, 42000 updates). Space, matter and the clock are born in this order, and the plane finally settles on one.

The clock — the time the system has of its own

As the system develops, it reaches a state in which **the whole turns by a constant angle without changing shape**. Once that happens, one can read from the state itself how many degrees it turned per update. **That is the tick of time for the system, and in this paper it is called the clock.**

Two conditions must hold for it to be readable: the overlap with the immediately preceding state must exceed a floor, and the amplitude of the component carrying the rotation must exceed a floor. The first update at which both hold is when the **clock is born**.

The count of updates kept from outside and this clock are different things. "The clock is born at the 2nd update" means that after two updates on the outside scale, a tick of time has appeared inside the system.

The condensate

At the location where the background oscillation sits, the phases of the gathered values align so that **squaring them and adding gives zero** (they cancel). This state is called the **condensate**. Particle-like waves are born on top of this lump. Here we say the condensate exists when the residue of that cancellation is below 10.

The third direction

The distribution of the background oscillation initially lies within a single plane. When a component growing outside that plane appears, a **third direction** stands up. The first update at which the fraction outside exceeds 5% is when **space is born**.

But "having grown outside" does not fix the direction. If two or more planes compete as the plane to grow out of, there is no telling which to take as reference. We measure separately the degree to which one plane is singled out, and when that exceeds a fixed value we say **the third direction is determined**.

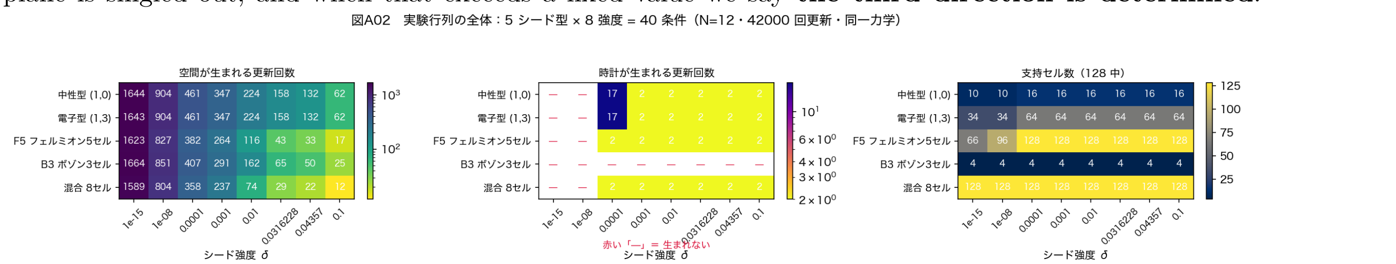

Figure 3The whole main experiment. 5 seed placements × 8 strengths = 40 conditions, run with identical dynamics, resolution and number of updates. A red dash means not born.

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1) The form of the seed gives partial control over which kind of particle is born

Keeping the winding number k of the first direction fixed, we compared **two seeds differing only in the winding number m of the second direction**, placed with the same strength. One targets a partner that is an uncharged wave (neutrino type); the other targets a partner that is an electron-type wave.

The number of updates at which inflation starts, and the number at which the clock is born, **agreed exactly for 7 of the 8 strengths**. They differed by a single update only at the weakest strength, where the run is indistinguishable from the no-seed case.

Even so, the locations occupied by the resulting waves split into 16 and 64. For the electron type the pattern is a checkerboard: when the first winding number is even so is the second, and when the first is odd so is the second.

That is, **"when it is born" and "what is born" are decided separately**. The former is insensitive to the seed placement; the latter is decided by it.

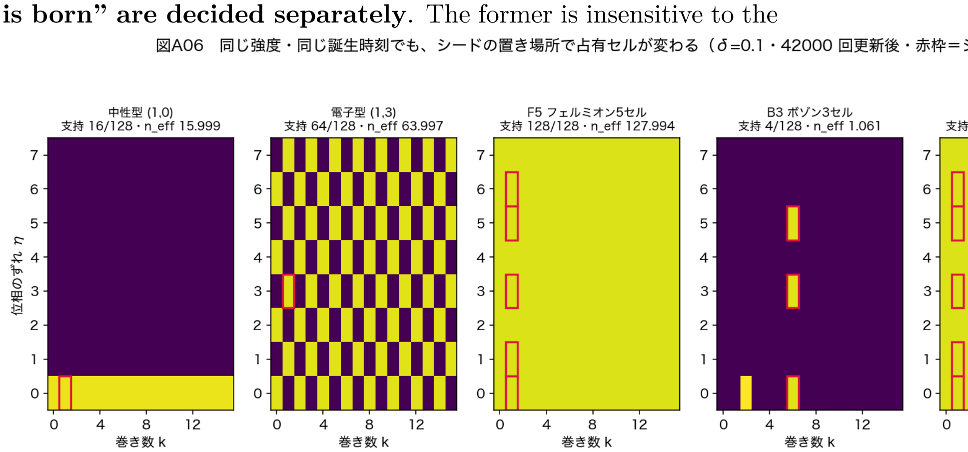

Figure 4Locations occupied, for five seed placements (strength 0.1, after 42000 updates). Red frames mark the seed. The neutrino type occupies 16, the electron type a checkerboard of 64, and the even-band-only seed just 4.

Being born as intended happens only at intermediate strength

Dividing the power at the targeted location by the power at non-targeted locations measures how concentrated the result is at the target.

Seed strength	10	10	**10^3**	10^2	0.03 and above
Concentration	0	0.009	**10**	2×10^3	0.08

If the seed is too strong the wave spreads over all 128 locations and the aim stops working. The aim works best near strength 10^3, where concentration at the target reaches ten thousand-fold; increasing the strength drops it by six orders of magnitude.

This concentration also varies by a factor of 120 with resolution (57 at N = 3, 0.47 at N = 20).

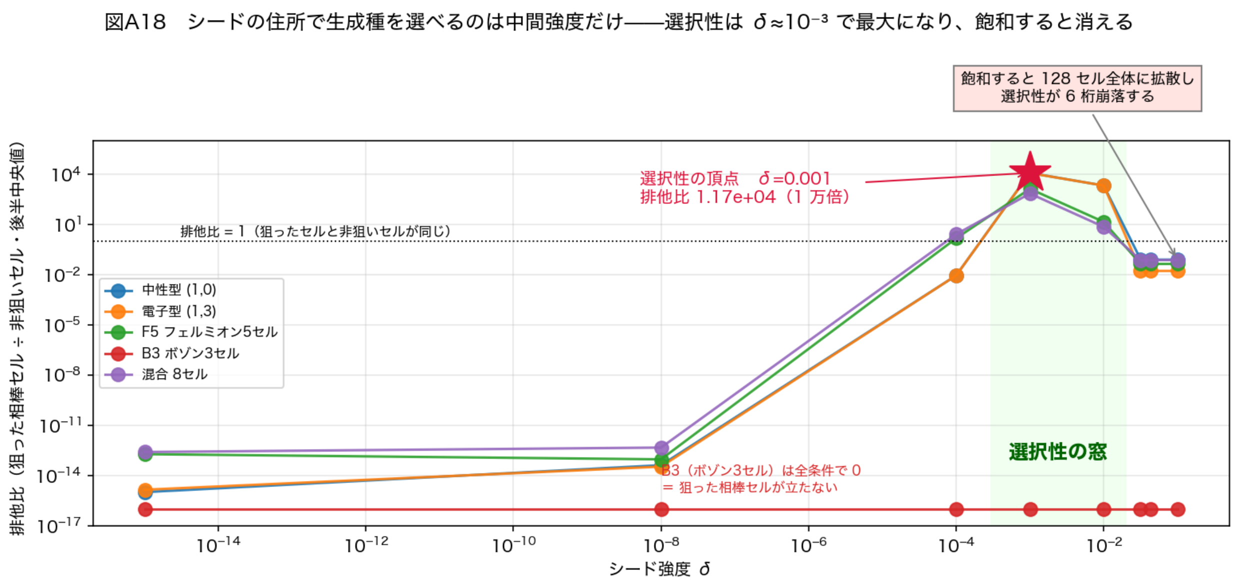

Figure 5Concentration at the targeted location peaks near strength 10^3 and falls by six orders of magnitude as the seed is strengthened.

Figure 6The map of 128 locations when concentrated (strength 10^3) and when fully spread (strength 0.1), for the same electron-type seed.

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2) The seed amplitude appears to have a window, but that appearance is created by the length of the observation

If one looks only as far as 42000 updates, it appears that too weak a seed yields inflation and three directions but no particle-like wave, that a stronger seed yields particle-like waves, and that strengthening further has no additional effect.

However, when the run was extended to 300000 updates, **at strength 10^2, where nothing appeared to happen within 42000 updates, particle-like waves rose all at once at update 262,751**. It was not that it had no effect; it took time to take effect.

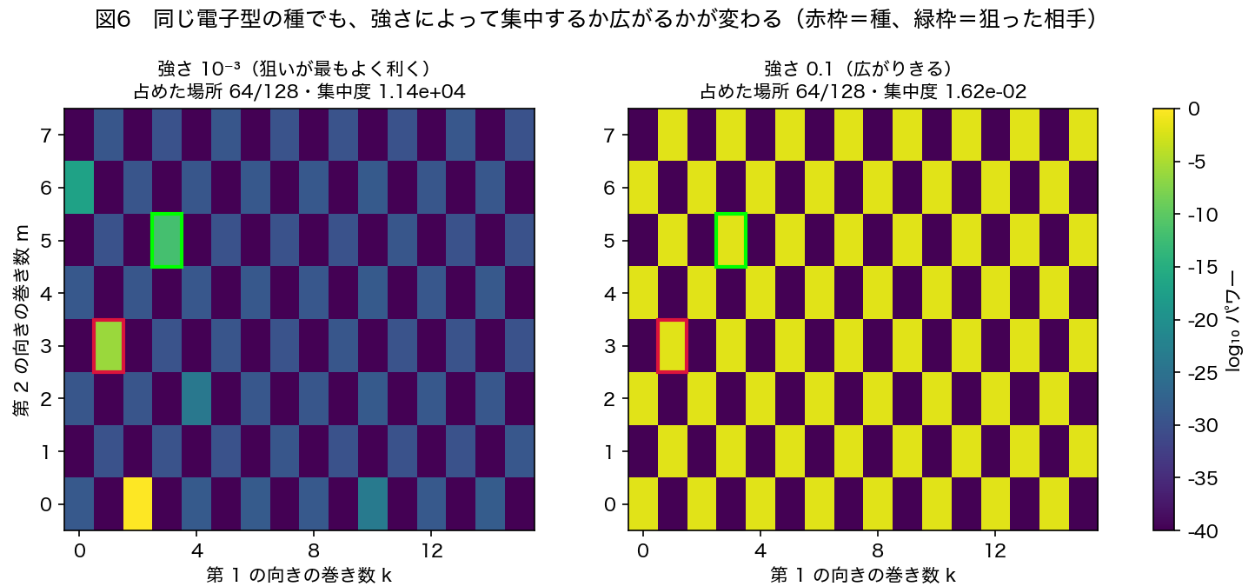


Figure 1: Figure 6The map of 128 locations when concentrated (strength 10^3) and when fully spread (strength 0.1), for the same electron-type seed.

Seed strength	10^3	$\mathbf{10^2}$	0.03	0.044	0.1
Updates until they finish appearing	not reached in 42000	**262,751**	25,760	13,391	2,352

Within the measured range no boundary strength appears; the data connect smoothly through the following empirical law.

```
(updates until they finish appearing)  (power placed on odd bands)^(1.073)
goodness of fit R^2 = 0.996 (14 points)
```

Extrapolating this law, the number of updates diverges as the power placed on odd bands approaches zero. **Therefore, seen within any finite observation time, an apparent lower bound must arise.** Indeed a seed placed on no odd band at all (power exactly zero) never finishes, at any strength.

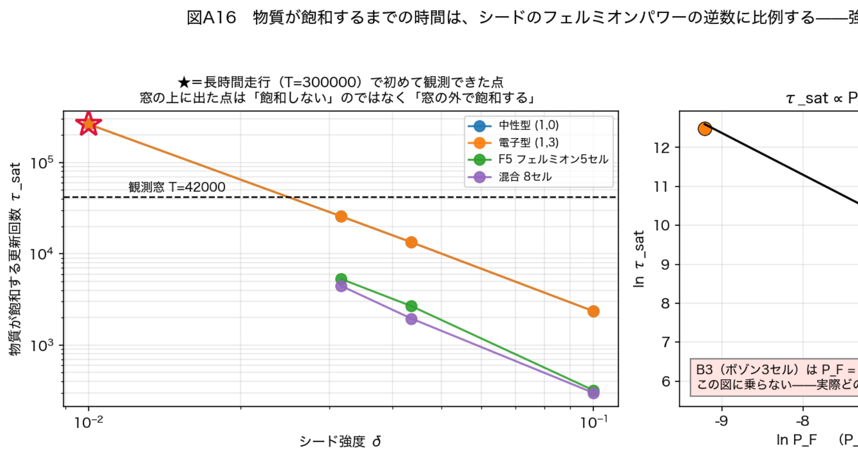
However, **it has not been shown that every non-zero seed necessarily finishes in the infinite-time limit.** What has been shown is that what looked like a threshold at 42000 updates collapsed at 300000, and that the 14 measurable points lie on this law.

Within the measured range, then, the strength of the seed does not decide whether particle-like waves can form, but how long they take to form.

The amount produced differs qualitatively before and after.

- Before, the amount on odd bands is **exactly what was put in** (the seed power itself)
- After, **odd and even bands are roughly half and half**

The lower bound “too weak a seed yields no particle-like wave” is likewise likely an artefact of the observation



time. A long run is needed to check this (not done).

Figure 7Updates until the particle-like waves finish appearing. Stars mark points that only the 300000-update runs could reach. Points above the 42000 line do not fail to finish; they finish outside that line. Right: the same data against the power placed on odd bands.

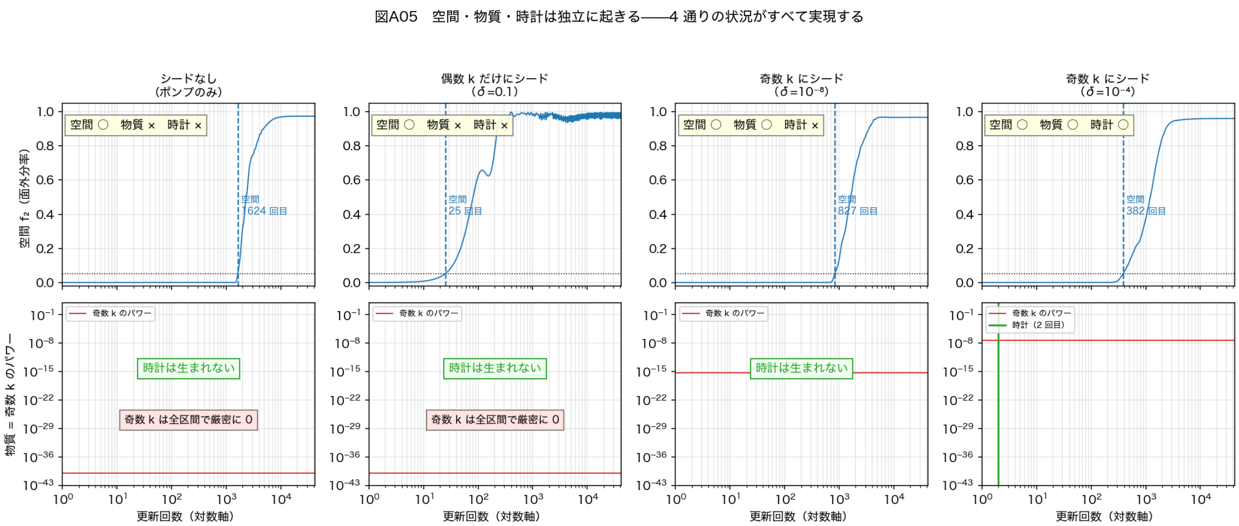


Figure 2: Figure 8Space, matter and the clock do not share a birth condition. All four combinations occur.

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3) The strength of the seed changes the run-up to the onset of inflation

The run-up shortens in proportion to the logarithm of the seed strength.

```
run-up = 9.892  48.611 × ln(magnitude of the seed summed as complex numbers)
goodness of fit R² = 0.99987
```

What matters here is **not the total power of the seed but the magnitude obtained by adding the seed as complex numbers**. Organised by total power instead, the scatter worsens by a factor of 7.7.

This follows from the update procedure. The first half of an update **looks only at the sum of the 128 values taken as complex numbers** and applies the same rotation to every location. Two initial states with the same sum are therefore indistinguishable to it.

This was tested directly. Without changing the seed's power or placement at all, the **phases were arranged to cancel**, so that only the sum becomes zero. It was then predicted, **before measuring**, that the result would agree — down to the number of updates — with the case having no seed on even bands. **All four strengths hit exactly.**

Seed strength	ordinary seed	phase-cancelled seed	no seed on even bands
0.01	74	**116**	116
0.03	29	**43**	43
0.044	22	**33**	33
0.1	12	**17**	17

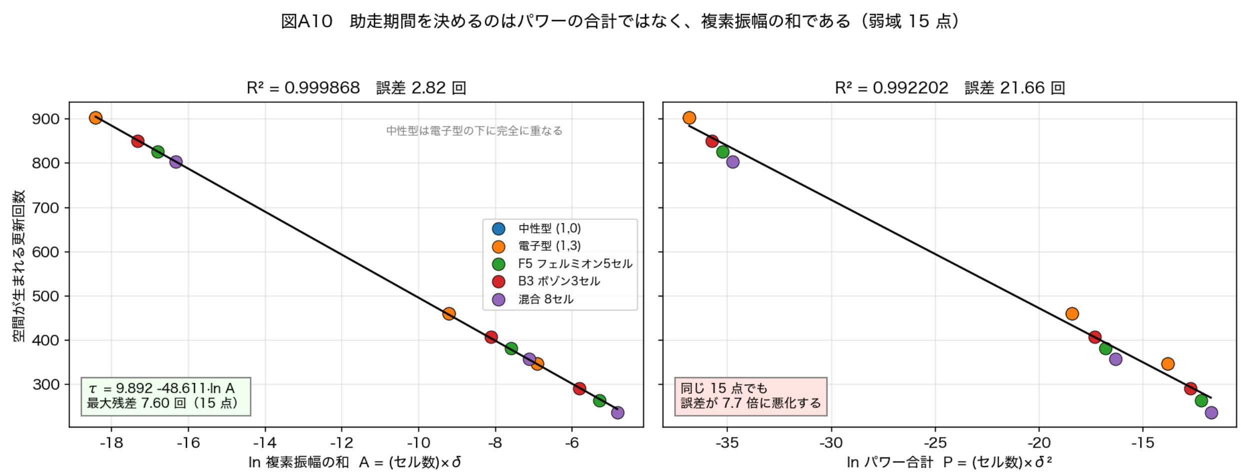

Figure 3: Figure 9The run-up. Left: organised by the seed summed as complex numbers. Right: the same 15 points organised by total power.

Figure 9The run-up. Left: organised by the seed summed as complex numbers. Right: the same 15 points organised by total power.

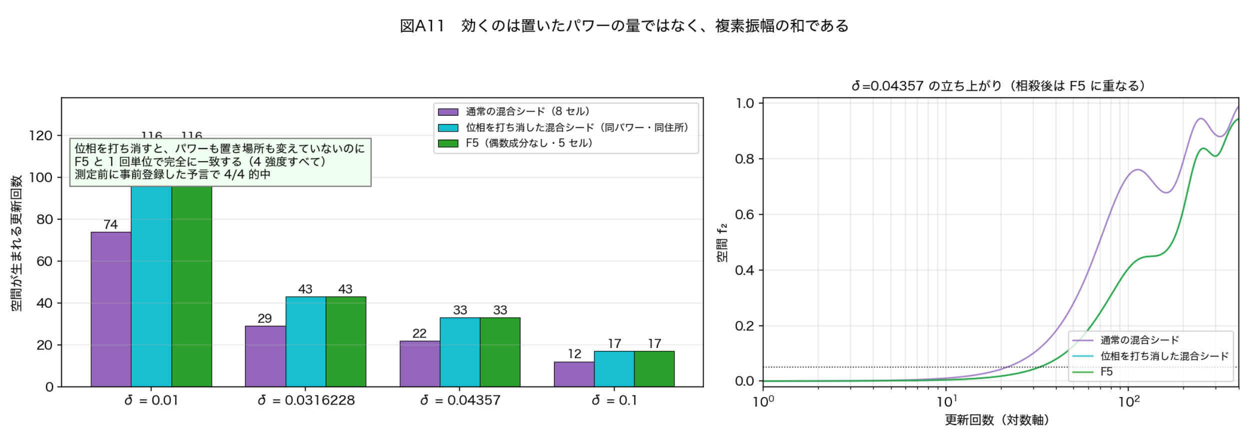

Figure 4: Figure 10The phase-cancellation control. Without changing power or placement, the result agrees update for update with the case having no seed on even bands.

Figure 10The phase-cancellation control. Without changing power or placement, the result agrees update for update with the case having no seed on even bands.

This control experiment alone cannot separate the sum from the sum of squares

The cancellation used a triple of equal amplitudes with phases 0, 2/3 and 4/3. That arrangement makes the complex sum zero, but **it also makes the sum of squares zero** (both of order 10^1).

This experiment alone therefore cannot tell whether the first half of an update reads the "sum" or the "sum of squares". The paper says it is the sum because the update procedure itself uses only the argument of the sum. The experiment is consistent with that, but it does not on its own single out the sum.

To separate them one needs an arrangement whose sum is zero but whose sum of squares is not (for example a pair with phase difference). That was not done here.

The complex sum matters only for the onset

The "updates until the waves finish appearing" of section 2 is instead decided by total power. Two different rules of decision coexist in the same system.

What is observed	Decided by	Quantity that matters
Onset of inflation	first half of an update	the **sum** taken as complex numbers
Time when particle-like waves finish appearing	second half	the total **power**

The first half looks only at the sum. The second is proportional to a coefficient built from the ratio of odd-band to even-band power. Hence the quantity that matters depends on which phenomenon is observed.

A seed placed only on even bands satisfies both at once. Its odd-band power is zero, so the finishing time is infinite (and indeed it never finishes at any strength), while its onset lies on the straight line of the complex sum.

図A17　同じ系に性質の違う二つの法則がある——見る事象が線形部と非線形部のどちらに属するかで、効く量が変わる

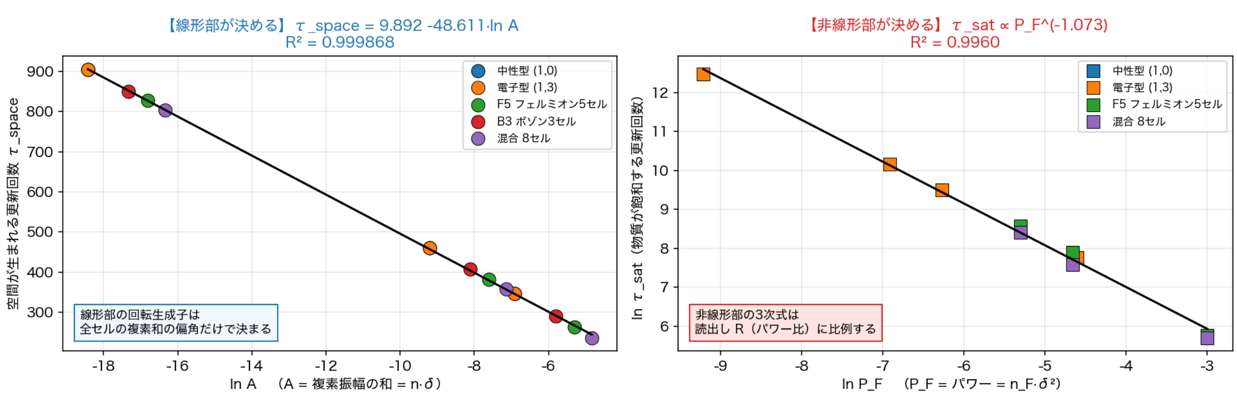

Figure 11Two rules of decision in one system. Left: the onset, decided by the first half of an update. Right: the finishing time, decided by the second half.

Figure 12Updates to onset for all 40 conditions. Re-plotted against the seed summed as complex numbers (right), the five placements fall on one curve.

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4) There is a minimum resolution: at N = 1, 2 neither inflation nor three directions appeared

At N = 1 there is not a single relation between points, and at N = 2 only one, so the system cannot be built at all.

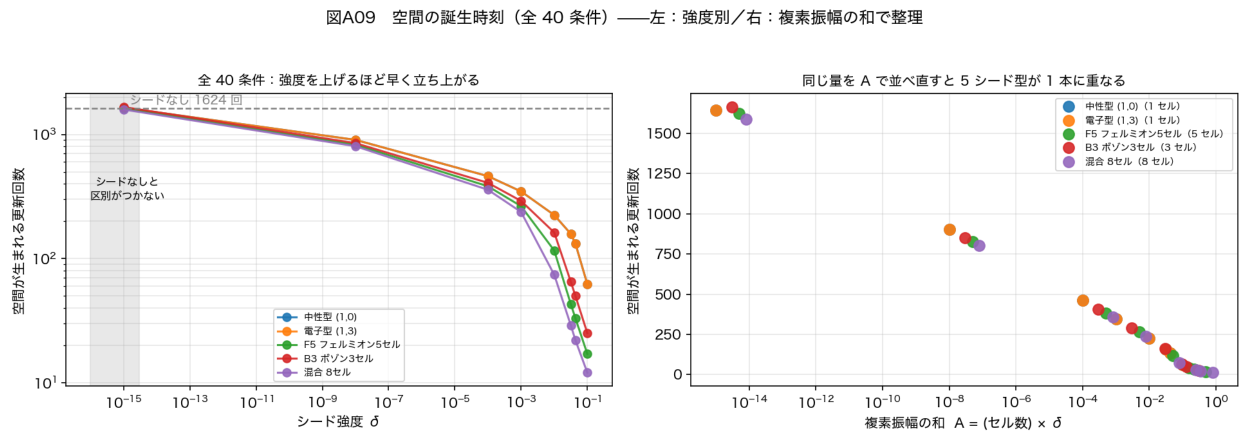

Figure 5: Figure 12Updates to onset for all 40 conditions. Re-plotted against the seed summed as complex numbers (right), the five placements fall on one curve.

There are singular resolutions: at N = 3, 4 the third direction is not determined

At N = 3 and N = 4 the planes compete and no single plane is selected, so the third direction is not determined. Three planes compete at N = 3 and two at N = 4. Moreover this competition **never settles; it keeps fluctuating over the whole run.** There is no steady state.

N = 5 and 6 show yet another aspect: **quiet stretches alternate with violently fluctuating ones.** Only from N = 7 upward does a single plane dominate.

At N = 8 and N = 10 the initial state itself could not be built. This, however, is because the implementation stops after one failed attempt; it has not been confirmed that it is structurally impossible (unverified).

Seeding acts to resolve the competition. Among the 16 resolutions that could be built, the number showing competition falls from 8 with no seed, to 4 with a seed at one location, to 2 with seeds at eight locations. **N = 3 and N = 4 alone cannot be resolved by any amount of seeding.**

There is no relation of the kind "resolutions that compete have longer run-ups". The two are separate properties.

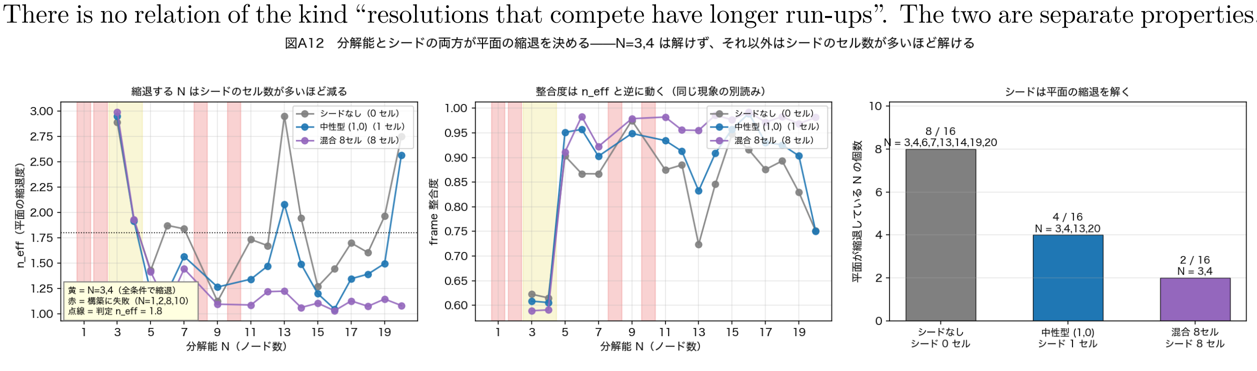

Figure 13Competition between planes, by resolution and seed. Yellow marks N = 3, 4 (competing under every condition); red marks resolutions where the initial state could not be built. Right: the more seed locations, the fewer resolutions compete.

Figure 14The judgements for four conditions over resolutions 1–20. In the no-seed rows only space is marked, with matter and the clock absent at every resolution.

Three anomalies that appear only at N = 4

1. Two planes compete and the third direction is not determined

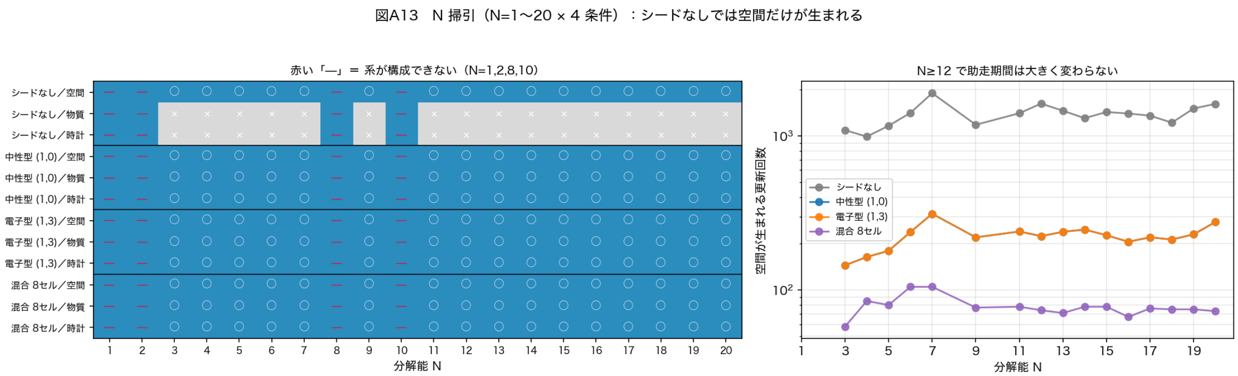

Figure 6: Figure 14The judgements for four conditions over resolutions 1–20. In the no-seed rows only space is marked, with matter and the clock absent at every resolution.

2. **The clock ticks 9.9 times faster than at other resolutions**
3. **It is the only resolution at which the readout crosses the special value 0.6972 reported in the previous paper**

Examining all 126 runs, the only ones to exceed that value are **the two runs with a single-location seed at N = 4**, reaching a maximum of 0.8019. The neutrino type and the electron type agree to 12 digits on that value. The closest approach is 0.0000068 (0.001% in relative terms).

But it only crosses; it does not stay. The readout lies within ±0.001 of that value for **only 9 of the 42000 updates**. At N = 4 the system keeps fluctuating violently after the waves have finished appearing, and the fluctuation happens to cross the value. At N = 12 the maximum is 0.6786, short of that value by 2.7%.

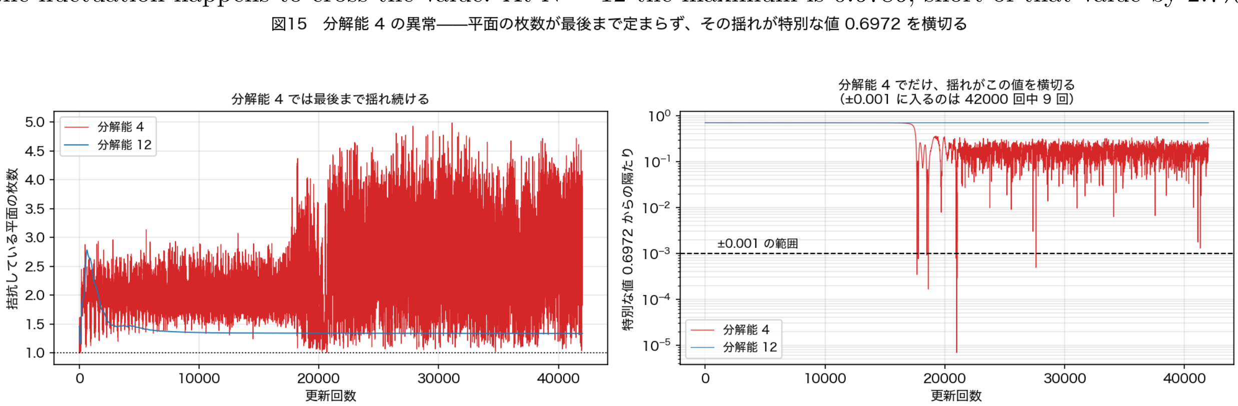

Figure 15The anomaly at resolution 4. The number of competing planes never settles (left), and that fluctuation crosses the special value 0.6972 (right).

From N 12 up to N = 20, tested, there was broadly no large change of behaviour

The run-up stays within 206–277 updates with no monotone trend.

There are two exceptions. Competition between planes reappears at N = 13 and N = 20. In particular **at N = 20 the third direction is not determined**.

After the waves finish appearing, the plane stops being determined again

“A single plane dominates from N = 7 upward” refers to the state before the particle-like waves finish appearing. Afterwards the number of competing planes keeps fluctuating at every resolution.

5) There are not four things that “are born” but six, and a strong seed destroys the last two

What the experiment judges is the following six.

Whether the system could be built / whether **space is born** (a third direction stands up) / whether **matter is born** (waves appear on odd bands) / whether **the clock is born** (the system can read its own time) / whether **the third direction is determined** / whether **the condensate exists**.

As the seed is strengthened, **the first four increase but the last two are lost**.

Seed strength	space	matter	clock	third direction determined	condensate
10^1, 10			×		
10, 10^3					
10^2					× (with many seed locations)
0.03 and above				×	×

A strong seed produces matter and a clock at the cost of the certainty of the third direction and of the condensate. Things do not simply come into being; there is a trade.

Furthermore, extending the run to 300000 updates, **the condensate is eventually destroyed even at strength 10^2**. That it looked intact at 42000 updates only means it had not yet been destroyed.

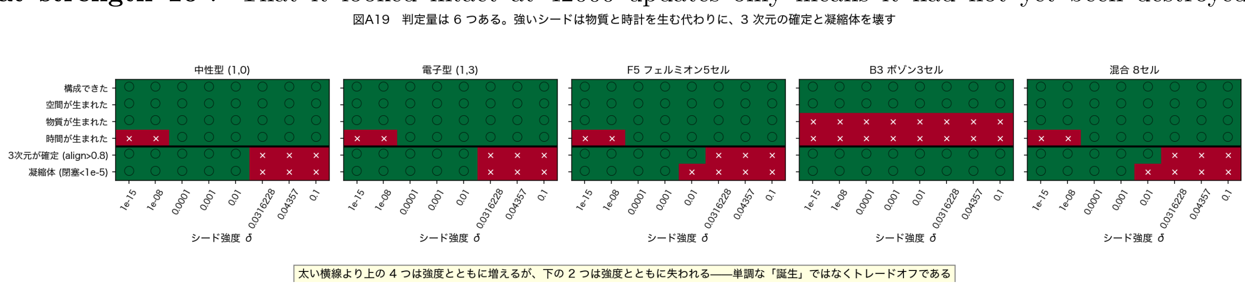

Figure 16The six criteria. Above the thick line the first four increase with seed strength; below it the last two are lost.

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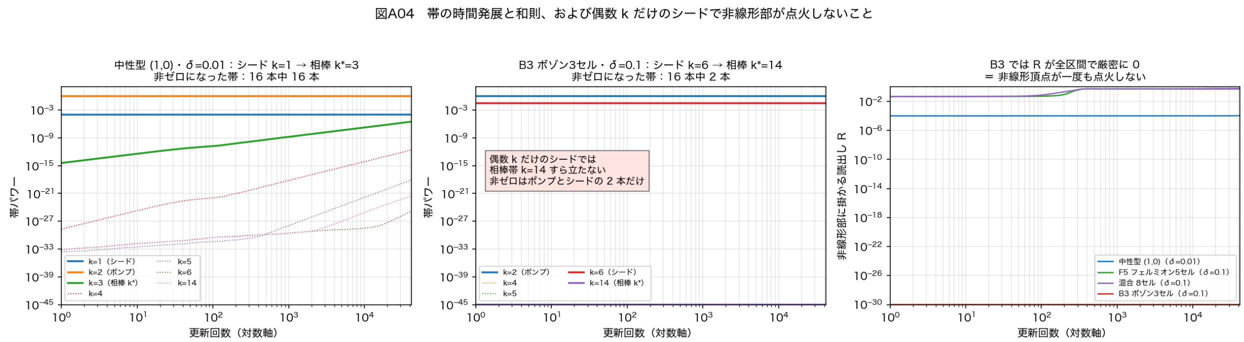
6) With a seed on even bands only, the interaction itself never occurs even once

With a seed on even bands only, **no wave whatsoever appears on the odd bands**, over the whole of 42000 updates and all 8 strengths. More than that, **only 2 of the 16 bands ever carry a wave** — the background oscillation and the seed itself — and **not even the partner band that this seed was aiming at stands up.**

The reason is this. The coefficient that sets the strength of the interaction is built from the ratio of odd-band to even-band power. If the odd bands are zero, that coefficient is zero. If the coefficient is zero, the whole interaction expression is zero. **With a seed on even bands only, the interaction never occurs and the system runs on the first half of the update alone.**

Why no wave appears on the odd bands follows from the update procedure itself. The interaction has the form of three waves meeting, so whether the outgoing wave lies on an even or an odd band is decided by the sum of the parities of the three incoming ones. Gather only even ones and only even ones come out.

In exchange, this seed **preserves the certainty of the third direction and the condensate at every strength**. Nothing is destroyed because no interaction occurs. It is the clearest control for the trade described in



section 5.

Figure 17Growth by band. Left: the neutrino type, where the targeted partner band stands up. Middle: the even-band-only seed, where only 2 bands ever carry a wave. Right: the coefficient setting the interaction strength, identically zero for the even-band-only seed.

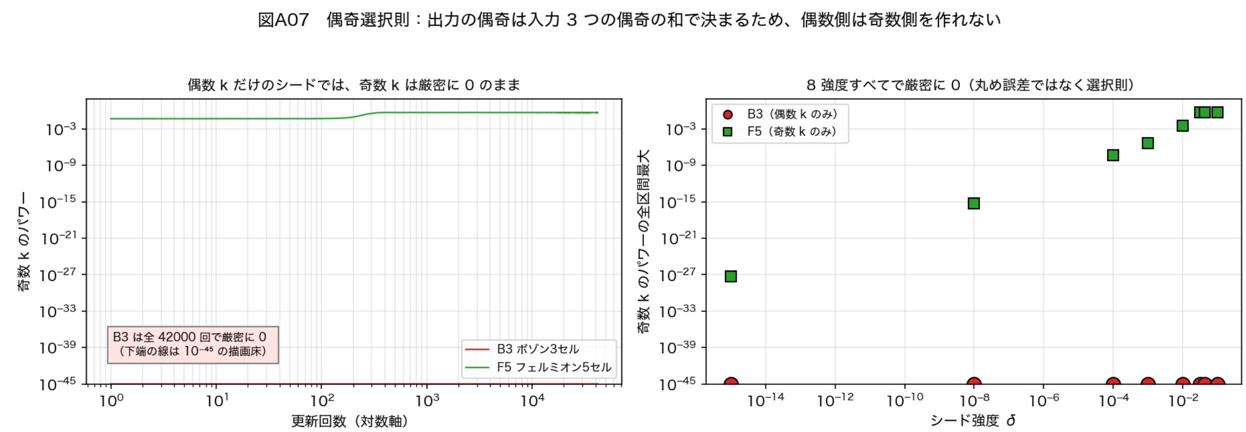


Figure 7: Figure 18With a seed on even bands only, the odd bands stay exactly zero over the whole of 42000 updates and all 8 strengths.

Figure 18With a seed on even bands only, the odd bands stay exactly zero over the whole of 42000 updates and all 8 strengths.

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7) Time appears twice in this system: the order of updates is not time

Two things that could be called time appear here. One is the **count of updates kept from outside**, which is merely a serial number we attach in order to run the experiment; it is not inside the system. The other is the **clock the system reads from its own state**, which exists only once the whole turns by a constant angle without changing shape.

These two must not be conflated. No quantity named time is fed into the dynamics of this system; what is fed in is only the update procedure. Nevertheless, as the system develops, a tick of rotation becomes readable from the relations among states. **Time is not something given, but something that became readable.**

Having a clock, its being constant, and its rate are three different things

The number of updates at which the system becomes able to read its own time, and the number at which the rate it reads settles to a constant value, are different quantities.

- With seeds at eight locations and 4000 updates, **the rate does not settle for N 15** (the time has become readable)
- Applying the phase cancellation of section 3, **the number of updates to settle falls from 15638 to 18, a factor of 870** (at strength 0.1, from 96 to 6)
- The rate of the clock itself varies with condition. Of 132 runs, **67 deviated from the reference by more than 5%**. The largest is the factor 9.9 at N = 4

That the clock has been born does not mean the system keeps a constant time.

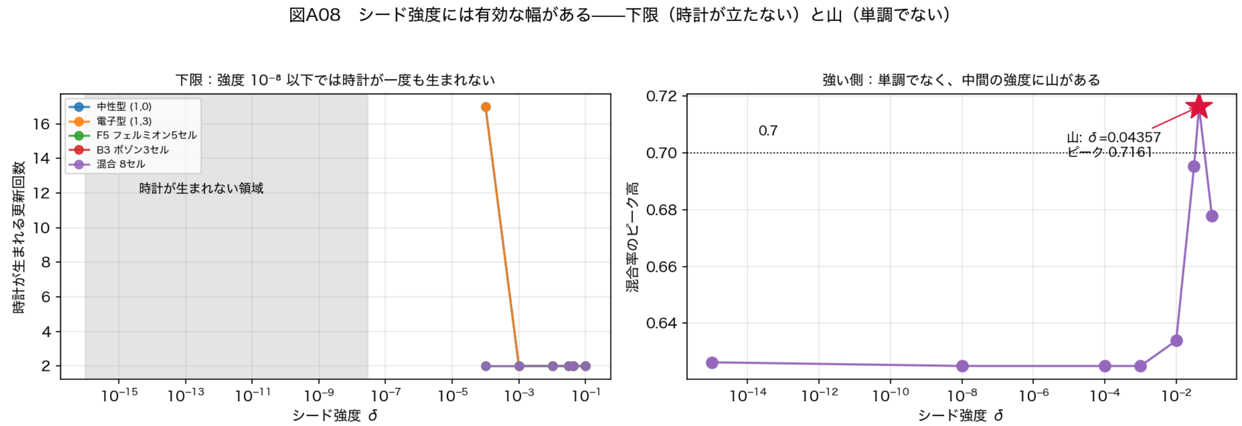

Figure 19Updates at which the clock is born (left) and the peak of the mixing ratio (right). In the grey region on the left, no clock is born within the 42000 updates observed.

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8) The classification by greatest common divisor found in the previous paper survives a change in the number of divisions of the period

The previous paper found that, inside the background sea, the properties of a wave depend not on the winding number of the second direction itself but on **the greatest common divisor of that winding number and the "number of divisions of the period"**.

By the number of divisions of the period is meant, as in the description of the system above, into how many parts one lap of the second direction is divided. A winding number has meaning only as a remainder modulo that number, so changing it should change which winding numbers fall into the same class. The previous paper registered this as a question to be tested. This paper tested it.

Re-running at the same number of divisions, 16, **all 62 values agreed bit for bit** with the stored results of the previous paper. Divisions of 8, 12 and 32 were then added, and **all 278 pairs came out as predicted, with zero failures**.

The decisive case is the pair of winding numbers 1 and 3. If the greatest common divisor governs, this pair should fall in the same class at divisions 8, 16 and 32, and in **different classes at 12 alone**. Measurement gives **agreement to 14 digits** at 8, 16 and 32, and a **60% difference at 12 alone**.

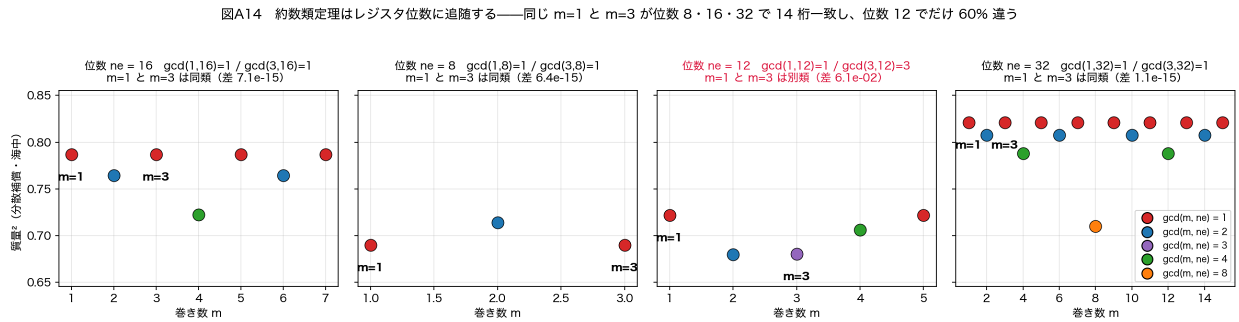

Figure 20The test of changing the number of divisions of the period. Points of the same colour line up at the same height. Winding numbers 1 and 3 agree to 14 digits at divisions 8, 16 and 32, and differ by 60% at 12 alone.

Open questions

1. Whether the failure to build the initial state at N = 8 and N = 10 is structural or an artefact of the implementation cutting off after one attempt
2. Whether "too weak a seed yields no particle-like wave" is likewise an artefact of the observation time
3. The long run at strength 10^2 with many seed locations. The saturation law (Figure 7) is drawn from 14 points; with those 2 added it becomes 16. The exponent may shift, but the conclusion that a finite observation time must produce an apparent lower bound does not
4. Whether the classification by greatest common divisor also holds in the many-body system (section 8 was verified in the two-wave system)
5. A test changing the number of divisions of the first direction (16 in this experiment)
6. Measurement of the coefficient the interaction actually uses (the relation to the value 0.6972 of the previous paper is decided there)
7. Matching the 62 kinds of the previous paper against which location in this experiment corresponds to which particle

The following go beyond the scope of this paper and are deferred to a sequel.

8. **An orthogonal experiment separating the run-up law from the saturation law.** The phase-cancellation experiment here fixed power and placement and varied the complex sum. Its orthogonal counterpart would **fix the sum and vary only the power**; the onset should be unchanged and only the finishing time should move. The same experiment would settle the "sum versus sum of squares" question of section 3, by including an arrangement whose sum is zero but whose sum of squares is not

9. **Falsification of the saturation law by extrapolation.** Freeze the exponent at the value from the existing 14 points, predict the finishing time for unmeasured strengths in advance, and go for the hit without refitting

10. **A formal audit of how far the system closes on relations alone.** For the three readouts used here (the plane and the third direction; bands and mixing ratio; the clock criterion), classify whether the inputs close on sets of relations alone, or whether quantities attached to individual points have crept in.

    This must be taken in two stages. Even if the readouts are confirmed to close on relations alone, that shows only that **the readout maps close on relations**, not that **the system itself runs on relations alone**. For the latter, the same audit is required of the construction of the initial state, the first half of the update, and the interaction.

    That is, one must check in turn, for each stage — building the state, updating, interacting, reading out — whether any quantity attached to a point itself has slipped in. This paper has not verified even the last of those stages.

AppendixFull list of runs, figures and programs

This appendix was generated automatically by make_paperA_appendix_v1.py, which scans the folder. Nothing in it was transcribed by hand. The Japanese edition carries the identical tables; only the headings are translated here.

Appendix AThe runs

126 stored run files (NPZ), 1174 MB in total. Runs differing in seed placement, seed strength, number of updates or resolution are counted as different runs.

- 40 conditions of the main experiment: 5 seed placements × 8 strengths (resolution 12, 42000 updates), all present
- Resolution sweep N = 1…20 for 4 conditions (no seed, neutrino type, electron type, mixed), 4000 updates
- Reproduction controls, phase-cancelled arms and their replicate, resolution-4 runs, and long runs at 300000 updates

Appendix BThe figures

20 figures in the body; 462 per-run record figures; 41 contact sheets used to inspect all of them. Naming is fig_<seed placement>[_T<updates>][_d<strength>][_rep-<tag>]_<family>[_N<resolution>]_v2.png, so any figure cited in the text is identified uniquely by its file name.

Family	Content	Count
4panel	space / number of competing planes / clock / cancellation residue	126
mix	power per band and mixing ratio	126
ledger	map of the 128 locations, and growth at the targeted one	78
summary	per-resolution summary	66
birth_matrix	the six criteria	66

Appendix CThe programs

Thirteen programs, plus five imported read-only. SHA-256 digests of every one are listed in the Japanese edition and in A__v1.json.

Role	File
Main experiment	run_nsweep_three_series_v2.py, run_tb_nsweep_1to20_v1.py
Additional seed placements	run_missing_seed_sweeps_T42000_v1.py
Phase-cancelled seed	run_phase_balanced_mixed_v1.py, run_phase_balanced_mixed_grid_v1.py
Test of the number of divisions	run_divisor_class_register_order_v1.py
Recomputation and cross-check of every claimed number	aggregate_paperA_claims_v1.py

Role	File
Figures	`make_paperA_figures_v2.py`, `make_paperA_figures_audit_v1.py`
Closest approach to 0.6972 over all runs	`probe_alpha_root_closest_approach_v1.py`
Contact sheets for full visual inspection	`make_contact_sheets_v1.py`
This appendix	`make_paperA_appendix_v1.py`
Long runs and resolution-4 runs	`run_stage4_longtime_orchestrator_v1.py`

The dynamics itself (interaction, plane and third-direction readout, band and mixing readout, clock criterion) and the two-wave map used in section 8 are imported read-only and never modified. Each run verifies their digests before and after.

Appendix DPre-registrations, reports and records

Pre-registration documents (predictions fixed before running), result reports, the full-figure audit record, and the JSON files holding every number.

Appendix EResults of the cross-check

Every number claimed was recomputed independently from the stored data and checked mechanically against the existing records.

Check	Result
Onset and clock counts recomputed from stored data vs. the records	40 conditions, 0 mismatches
Number of locations occupied	agrees
Odd bands exactly zero for the even-band-only seed	maximum 0.0
Run-up law	9.8922 48.6108 × ln(complex sum), R^2 = 0.999868
Same points organised by total power	R^2 = 0.992202
The four phase-cancelled conditions	all as predicted
Reproduction of the previous paper (16 divisions)	62 values, largest discrepancy 0.0e+00
Test of changing the number of divisions	278 pairs, 0 failures
Law for the finishing time	(odd-band power)^(1.073), R^2 = 0.9960 (14 points)